* Types of Neural Network

1/ Multi layer Perceptron (MLP):

-> 1 input, 1 output and in the middle there will be some hidden layers.
-> used at sueprvised problems.

2/ Convolutional Neural Network (CNN):

-> It is a special neural network where 1 layer would be the Convolution layer.
-> Mostly used on image and video processing.

3/ Recurrent Neural Network (RNN):

-> MLP and CNN is feed forward network. Information travels through first layer to last layer linearly.
-> But in RNN, In the hidden layer the information can go backward to improve performance.
-> Mostly used on NLP applications.

4/ Auto Encoders:

-> It is used to compress the information without losing the quality.

5/ Generative Adverserial Networks (GAN):

-> It is concept where you can make the program imagine.
-> For example, you can generate image, audio, text that doesn't exist, those are totally imagination
-> It this networks there is a genrator that generate those imaginated images and there is a discriminator that tells that the image is fake of real. If the genetor can fool the discriminator then we got a new imaginary picture.

History of Deep Learning:

-> Chapter-1 1960, the perceptron trick invented.
-> Chapter-2 1969 the first AI winter, there is a issue found on perceptron trick. Percepron can't learn the XOR function means it won't work on non linear function. Someone publish a paper on this topic and the trend of AI was dropped.
-> Chapter-3 1980 the rise, at 1986 a peper was published by the name of "learning representation using backpropagating errors". That says, add more perceptron layer (now known as hidden layer) to make it learn non-linear functions. Acturally create a neural network and train it with backpropagation.
-> Chapter-4 the 90's, the second AI winter. There was no

technique to intialize the weight while starting the netword and that causes errors.
-> Chapter-5, introduced unsupervised pre-training to initialize the weights.
-> Chapter-6 2012 the final rise, After this time it is now untoucable. Many research happened, everything improved and the trend is always high.